

Latency-Aware Unbeatable Rock-Paper-Scissors: Early Hand-Motion Prediction from Real-Time Hand Landmarks

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Team and Team Members

- Cho, MinKyu: model design, real-time inference pipeline, latency and win-rate evaluation, and final demo integration.
- Kim, Seunghwan: dataset collection protocol, video and landmark dataset construction, model training experiments, and dataset quality checks.

Introduction

Rock-paper-scissors is normally treated as a fair simultaneous game. This project asks what happens if an AI system can observe the user's hand motion, predict the final gesture before it is fully formed, and immediately display the winning counter-move. The project builds on a previous Teachable Machine rock-paper-scissors assignment, where a Keras image model classified a completed hand sign from a webcam frame and the computer selected a move randomly. That baseline is useful for recognition, but it is too late for winning: it waits until the final pose is visible.

Our project reframes rock-paper-scissors as a timing-sensitive embodied AI problem. The system will process live webcam frames, track the user's hand landmarks, predict whether the motion is becoming rock, paper, or scissors, and display the AI move that beats the prediction. The system will only act when confidence is high enough; otherwise it will abstain. In this proposal, "unbeatable" is the demo concept, while the scientific claim is measured performance: early prediction accuracy, end-to-end latency, abstention rate, and live win rate under a controlled protocol.

The project is worthwhile because it connects perception, temporal prediction, confidence gating, and latency-aware action selection in a compact real-time demo. A simple game makes the behavior easy to understand, but the technical challenge is nontrivial: winning requires predicting partial motion, not merely recognizing a completed hand pose.

Related Work

The core perception component is motivated by MediaPipe Hands, which provides real-time 21-point hand landmark tracking from a single RGB camera [1]. MediaPipe's current gesture recognition tools also provide predefined classes such as Closed_Fist, Open_Palm, and Victory, which map naturally to rock, paper, and scissors [2]. These systems make it possible to avoid training a detector from scratch and focus the project on timing and sequence prediction.

On-device hand gesture recognition research shows that landmark-based gesture classification can run in real time [3]. Work on custom hand gestures further suggests that small task-specific datasets can adapt pretrained hand representations to new gesture sets [4]. This supports our plan to use MediaPipe landmarks as the input and train a lightweight custom temporal model for the rock-paper-scissors setting.

The project is also inspired by high-speed janken robot systems that use fast visual sensing and actuation to beat a human player [5,6]. We do not attempt to reproduce that hardware. Instead, we adapt the idea into a software-only webcam demo: the AI response is rendered on screen, and the key question becomes whether the system can predict and respond fast enough to create a measurable advantage.

Prior work has also studied rock-paper-scissors as a computer vision task. Hunter built an RPS game using YOLOv3 gesture detection and opponent move prediction [7], while Xie et al. proposed an event-camera RPS recognition system for low-latency recognition [8]. More broadly, dynamic hand gesture recognition work shows that skeleton and landmark sequences can be modeled with recurrent networks such as LSTM or GRU models [9-12]. This literature motivates our use of a short sliding landmark window rather than a single image.

Method

We will compare static recognition against temporal prediction. The first baseline is the previous Teachable Machine image classifier, which predicts rock, paper, or scissors from a completed webcam frame. The second baseline is MediaPipe Gesture Recognizer, using Closed_Fist as rock, Open_Palm as paper, and Victory as scissors.

The proposed system uses MediaPipe Hand Landmarker to extract 21 hand landmarks from each webcam frame. Landmarks will be normalized around the wrist and scaled by palm size so that the model is less sensitive to camera distance and hand position. The feature vector will include normalized x, y, z landmark coordinates and first-order velocity features. A sliding window of recent frames will be passed to a small 2-layer GRU classifier with four output classes: rock, paper, scissors, and neutral/transition.

The neutral/transition class prevents the system from forcing a move during ambiguous motion. At inference time, the AI displays a counter-move only when the predicted class is rock, paper, or scissors, the confidence exceeds a threshold, and the prediction is stable for 2-3 frames. The counter-move policy is deterministic: predicted rock maps to AI paper, predicted paper maps to AI scissors, and predicted scissors maps to AI rock. If confidence is low, the app waits and logs the abstention.

The dataset will consist of short webcam clips that begin from a neutral ready pose, move at normal RPS speed, and briefly hold the final sign. Our first target is 3-5 participants, 30 clips per class per participant, and four classes: rock, paper, scissors, and neutral/transition. We will keep raw videos, extracted landmark sequences, train/validation/test splits, trained model files, and live demo logs. When enough participants are available, the split will be participant-aware to test generalization across hand shapes and throwing styles.

Evaluation will report final gesture accuracy, early prediction accuracy from partial motion, decision latency from frame capture to rendered AI response, live win rate in controlled rounds, and abstention rate. This separates recognition quality from timing behavior and prevents confidence gating from hiding failure cases.

Timeline

Before Progress Update 1, we will finalize the literature review, define the data collection protocol, implement MediaPipe landmark extraction, and run the existing Teachable Machine baseline.

Before Progress Update 2, we will collect the first dataset, train an initial GRU model, compare it with the static baselines, and build a live overlay prototype showing landmarks, prediction, confidence, latency, and AI counter-move.

Before the final submission, we will tune thresholds, evaluate latency and live win rate, record a demo video, write the final report, and prepare presentation slides.

Expected Deliverables

The main project deliverable is a real-time webcam application that displays tracked hand landmarks, predicted gesture, confidence score, measured latency, and the AI's winning counter-move. Supporting deliverables include a trained landmark-window GRU classifier, dataset collection notes, evaluation logs, a demo video, and a final comparison against static baselines. The v1 scope is software-only; no physical robot hand is required.

Deliverables

- proposal.pdf
- presentation-slides.pdf

References

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